

The network layer is concerned with _____ of data.

- a) bits
- b) frames
- c) packets
- d) bytes

[View Answer](#)

Answer: c

Which one of the following is not a function of network layer?

- a) routing
- b) inter-networking
- c) congestion control
- d) error control

[View Answer](#)

Answer: d

A 4 byte IP address consists of _____

- a) only network address
- b) only host address
- c) network address & host address
- d) network address & MAC address

[View Answer](#)

Answer: c

. Which of the following routing algorithms can be used for network layer design?

- a) shortest path algorithm
- b) distance vector routing
- c) link state routing
- d) all of the mentioned

[View Answer](#)

Answer: d

. Which of the following routing algorithms can be used for network layer design?

- a) shortest path algorithm

- b) distance vector routing
- c) link state routing
- d) all of the mentioned

[View Answer](#)

Answer: d

Which one of the following algorithm is not used for congestion control?

- a) traffic aware routing
- b) admission control
- c) load shedding
- d) routing information protocol

[View Answer](#)

Answer: d

The network layer protocol for internet is _____

- a) ethernet
- b) internet protocol
- c) hypertext transfer protocol
- d) file transfer protocol

[View Answer](#)

Answer: b

ICMP is primarily used for _____

- a) error and diagnostic functions
- b) addressing
- c) forwarding
- d) routing

[View Answer](#)

Answer: a

Transport layer aggregates data from different applications into a single stream before passing it to _____

- a) network layer
- b) data link layer

- c) application layer
- d) physical layer

[View Answer](#)

Answer: a

Which of the following are transport layer protocols used in networking?

- a) TCP and FTP
- b) UDP and HTTP
- c) TCP and UDP
- d) HTTP and FTP

[View Answer](#)

Answer: c

User datagram protocol is called connectionless because _____

- a) all UDP packets are treated independently by transport layer
- b) it sends data as a stream of related packets
- c) it is received in the same order as sent order
- d) it sends data very quickly

[View Answer](#)

Answer: a

Transmission control protocol _____

- a) is a connection-oriented protocol
- b) uses a three way handshake to establish a connection
- c) receives data from application as a single stream
- d) all of the mentioned

[View Answer](#)

Answer: d

An endpoint of an inter-process communication flow across a computer network is called

- a) socket
- b) pipe
- c) port

d) machine

[View Answer](#)

Answer: a

Which one of the following is a version of UDP with congestion control?

a) datagram congestion control protocol

b) stream control transmission protocol

c) structured stream transport

d) user congestion control protocol

[View Answer](#)

Answer: a

A _____ is a TCP name for a transport service access point.

a) port

b) pipe

c) node

d) protocol

[View Answer](#)

Answer: a

. Transport layer protocols deals with _____

a) application to application communication

b) process to process communication

c) node to node communication

d) man to man communication

[View Answer](#)

Answer: b

. Which of the following is a transport layer protocol?

a) stream control transmission protocol

b) internet control message protocol

c) neighbor discovery protocol

d) dynamic host configuration protocol

[View Answer](#)

Answer: a

Physical or logical arrangement of network is _____

- a) Topology
- b) Routing
- c) Networking
- d) Control

[View Answer](#)

Answer: a

. Which network topology requires a central controller or hub?

- a) Star
- b) Mesh
- c) Ring
- d) Bus

[View Answer](#)

Answer: a

. _____ topology requires a multipoint connection.

- a) Star
- b) Mesh
- c) Ring
- d) Bus

[View Answer](#)

Answer: d

Data communication system spanning states, countries, or the whole world is _____

- a) LAN
- b) WAN
- c) MAN
- d) PAN

[View Answer](#)

Answer: b

. Data communication system within a building or campus is _____

- a) LAN
- b) WAN
- c) MAN
- d) PAN

[View Answer](#)

Answer: a

WAN stands for _____

- a) World area network
- b) Wide area network
- c) Web area network
- d) Web access network

[View Answer](#)

Answer: b

In TDM, slots are further divided into _____

- a) Seconds
- b) Frames
- c) Packets
- d) Bits

[View Answer](#)

Answer: b

. _____ is the multiplexing technique that shifts each signal to a different carrier frequency.

- a) FDM
- b) TDM
- c) Both FDM & TDM
- d) PDM

[View Answer](#)

Answer: a

The sharing of a medium and its link by two or more devices is called _____

- a) Fully duplexing

- b) Multiplexing
- c) Microplexing
- d) Duplexing

[View Answer](#)

Answer: b

Multiplexing is used in _____

- a) Packet switching
- b) Circuit switching
- c) Data switching
- d) Packet & Circuit switching

[View Answer](#)

Answer: b

Which multiplexing technique used to transmit digital signals?

- a) FDM
- b) TDM
- c) WDM
- d) FDM & WDM

[View Answer](#)

Answer: b

If there are n signal sources of same data rate, then the TDM link has _____ slots.

- a) n
- b) $n/2$
- c) $n*2$
- d) 2^n

[View Answer](#)

Answer: a

If link transmits 4000 frames per second, and each slot has 8 bits, the transmission rate of circuit this TDM is _____

- a) 32kbps
- b) 500bps

c) 500kbps

d) 32bps

[View Answer](#)

Answer: a

The state when dedicated signals are idle are called _____

a) Death period

b) Poison period

c) Silent period

d) Stop period

[View Answer](#)

Answer: c

Multiplexing provides _____

a) Efficiency

b) Privacy

c) Anti jamming

d) Both Efficiency & Privacy

[View Answer](#)

Answer: d

In TDM, the transmission rate of a multiplexed path is always _____ the sum of the transmission rates of the signal sources.

a) Greater than

b) Lesser than

c) Equal to

d) Equal to or greater than

[View Answer](#)

Answer: a

In TDM, slots are further divided into _____

a) Seconds

b) Frames

- c) Packets
- d) Bits

Which of the following delay is faced by the packet in travelling from one end system to another?

- a) Propagation delay
- b) Queuing delay
- c) Transmission delay
- d) All of the mentioned

[View Answer](#)

Answer: d

For a 10Mbps Ethernet link, if the length of the packet is 32bits, the transmission delay is _____ (in microseconds)

- a) 3.2
- b) 32
- c) 0.32
- d) 320

[View Answer](#)

Answer: a

The time required to examine the packet's header and determine where to direct the packet is part of _____

- a) Processing delay
- b) Queuing delay
- c) Transmission delay
- d) Propagation delay

[View Answer](#)

Answer: a

Given L = number of bits in the packet, a = average rate and R = transmission rate. The Traffic intensity in the network is given by _____

- a) La/R
- b) LR/a
- c) R/La

d) Ra/L

[View Answer](#)

Answer: a

In the transfer of file between server and client, if the transmission rates along the path is 10Mbps, 20Mbps, 30Mbps, 40Mbps. The throughput is usually _____

a) 20Mbps

b) 10Mbps

c) 40Mbps

d) 50Mbps

[View Answer](#)

Answer: b

If end to end delay is given by $d_{end-end} = N(d_{proc} + d_{trans} + d_{prop})$ is a non congested network. The number of routers between source and destination is?

a) $N/2$

b) N

c) $N-1$

d) $2N$

[View Answer](#)

Answer: c

The total nodal delay is given by _____

a) $d_{nodal} = d_{proc} - d_{queue} + d_{trans} + d_{prop}$

b) $d_{nodal} = d_{proc} + d_{trans} - d_{queue}$

c) $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

d) $d_{nodal} = d_{proc} + d_{queue} - d_{trans} - d_{prop}$

[View Answer](#)

Answer: c

. In a network, If P is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero?

a) Propagation delay

b) Queuing delay

- c) Transmission delay
- d) Processing delay

[View Answer](#)

Answer: b

Transmission delay does not depend on _____

- a) Packet length
- b) Distance between the routers
- c) Transmission rate
- d) Bandwidth of medium

[View Answer](#)

Answer: b

Propagation delay depends on _____

- a) Packet length
- b) Transmission rate
- c) Distance between the routers
- d) Speed of the CPU

[View Answer](#)

Answer: c

. FTP server listens for connection on port number _____

- a) 20
- b) 21
- c) 22
- d) 23

[View Answer](#)

Answer: b

In FTP protocol, client contacts server using _____ as the transport protocol.

- a) transmission control protocol
- b) user datagram protocol
- c) datagram congestion control protocol

d) stream control transmission protocol

[View Answer](#)

Answer: a

In Active mode FTP, the client initiates both the control and data connections.

a) True

b) False

[View Answer](#)

Answer: b

The File Transfer Protocol is built on _____

a) data centric architecture

b) service oriented architecture

c) client server architecture

d) connection oriented architecture

[View Answer](#)

Answer: c

In File Transfer Protocol, data transfer cannot be done in _____

a) stream mode

b) block mode

c) compressed mode

d) message mode

[View Answer](#)

Answer: d

. Expansion of FTP is _____

a) Fine Transfer Protocol

b) File Transfer Protocol

c) First Transfer Protocol

d) Fast Transfer Protocol

[View Answer](#)

Answer: b

FTP is built on _____ architecture.

- a) Client-server
- b) P2P
- c) Data centric
- d) Service oriented

[View Answer](#)

Answer: a

FTP uses _____ parallel TCP connections to transfer a file.

- a) 1
- b) 2
- c) 3
- d) 4

[View Answer](#)

Answer: b

. Identify the incorrect statement regarding FTP.

- a) FTP stands for File Transfer Protocol
- b) FTP uses two parallel TCP connections
- c) FTP sends its control information in-band
- d) FTP sends exactly one file over the data connection

[View Answer](#)

Answer: c

If 5 files are transferred from server A to client B in the same session. The number of TCP connections between A and B is _____

- a) 5
- b) 10
- c) 2
- d) 6

[View Answer](#)

Answer: d

FTP server _____

- a) Maintains state information
- b) Is stateless
- c) Has single TCP connection for a file transfer
- d) Has UDP connection for file transfer

[View Answer](#)

Answer: a

The commands, from client to server, and replies, from server to client, are sent across the control connection in _____ bit ASCII format.

- a) 8
- b) 7
- c) 3
- d) 5

[View Answer](#)

Answer: b

The data transfer mode of FTP, in which all the fragmenting has to be done by TCP is _____

- a) Stream mode
- b) Block mode
- c) Compressed mode
- d) Message mode

[View Answer](#)

Answer: a

The password is sent to the server using _____ command.

- a) PASSWD
- b) PASS
- c) PASSWORD
- d) PWORD

[View Answer](#)

Answer: b

. When the mail server sends mail to other mail servers it becomes _____

- a) SMTP server
- b) SMTP client
- c) Peer
- d) Master

[View Answer](#)

Answer: b

. If you have to send multimedia data over SMTP it has to be encoded into _____

- a) Binary
- b) Signal
- c) ASCII
- d) Hash

[View Answer](#)

Answer: c

Expansion of SMTP is _____

- a) Simple Mail Transfer Protocol
- b) Simple Message Transfer Protocol
- c) Simple Mail Transmission Protocol
- d) Simple Message Transmission Protocol

[View Answer](#)

Answer: a

. In SMTP, the command to write receiver's mail address is written with the command _____

- a) SEND TO
- b) RCPT TO
- c) MAIL TO
- d) RCVR TO

[View Answer](#)

Answer: b

The underlying Transport layer protocol used by SMTP is _____

- a) TCP
- b) UDP
- c) Either TCP or UDP
- d) IMAP

[View Answer](#)

Answer: a

Choose the statement which is wrong incase of SMTP?

- a) It requires message to be in 7bit ASCII format
- b) It is a pull protocol
- c) It transfers files from one mail server to another mail server
- d) SMTP is responsible for the transmission of the mail through the internet

[View Answer](#)

Answer: b

Internet mail places each object in _____

- a) Separate messages for each object
- b) One message
- c) Varies with number of objects
- d) Multiple messages for each object

[View Answer](#)

Answer: b

Typically the TCP port used by SMTP is _____

- a) 25
- b) 35
- c) 50
- d) 15

[View Answer](#)

Answer: a

A session may include _____

- a) Zero or more SMTP transactions

- b) Exactly one SMTP transactions
- c) Always more than one SMTP transactions
- d) Number of SMTP transactions cant be determined

[View Answer](#)

Answer: a

Which of the following is an example of user agents for e-mail?

- a) Microsoft Outlook
- b) Facebook
- c) Google
- d) Tumblr

[View Answer](#)

Answer: a

. When the sender and the receiver of an email are on different systems, we need only

- a) One MTA
- b) Two UAs
- c) Two UAs and one MTA
- d) Two UAs and two MTAs

[View Answer](#)

Answer: d

. User agent does not support this _____

- a) Composing messages
- b) Reading messages
- c) Replying messages
- d) Routing messages

[View Answer](#)

Answer: d

The entire hostname has a maximum of _____

- a) 255 characters
- b) 127 characters

- c) 63 characters
- d) 31 characters

[View Answer](#)

Answer: a

. A DNS client is called _____

- a) DNS updater
- b) DNS resolver
- c) DNS handler
- d) none of the mentioned

[View Answer](#)

Answer: b

Servers handle requests for other domains _____

- a) directly
- b) by contacting remote DNS server
- c) it is not possible
- d) none of the mentioned

[View Answer](#)

Answer: b

DNS database contains _____

- a) name server records
- b) hostname-to-address records
- c) hostname aliases
- d) all of the mentioned

[View Answer](#)

Answer: d

. If a server has no clue about where to find the address for a hostname then _____

- a) server asks to the root server
- b) server asks to its adjacent server
- c) request is not processed

d) none of the mentioned

[View Answer](#)

Answer: a

Which one of the following allows client to update their DNS entry as their IP address change?

a) dynamic DNS

b) mail transfer agent

c) authoritative name server

d) none of the mentioned

[View Answer](#)

Answer: a

The right to use a domain name is delegated by domain name registers which are accredited by _____

a) internet architecture board

b) internet society

c) internet research task force

d) internet corporation for assigned names and numbers

[View Answer](#)

Answer: d

The domain name system is maintained by _____

a) distributed database system

b) a single server

c) a single computer

d) none of the mentioned

[View Answer](#)

Answer: a

Which one of the following is not true?

a) multiple hostnames may correspond to a single IP address

b) a single hostname may correspond to many IP addresses

c) a single hostname may correspond to a single IP address

d) none of the mentioned

[View Answer](#)

Answer: c

Which of the following is false with respect to UDP?

- a) Connection-oriented
- b) Unreliable
- c) Transport layer protocol
- d) Low overhead

[View Answer](#)

Answer: a

Return value of the UDP port "Chargen" is _____

- a) String of characters
- b) String of integers
- c) Array of characters with integers
- d) Array of zero's and one's

[View Answer](#)

Answer: a

Beyond IP, UDP provides additional services such as _____

- a) Routing and switching
- b) Sending and receiving of packets
- c) Multiplexing and demultiplexing
- d) Demultiplexing and error checking

[View Answer](#)

Answer: d

What is the main advantage of UDP?

- a) More overload
- b) Reliable
- c) Low overhead
- d) Fast

[View Answer](#)

Answer: c

Port number used by Network Time Protocol (NTP) with UDP is _____

- a) 161
- b) 123
- c) 162
- d) 124

[View Answer](#)

Answer: b

What is the header size of a UDP packet?

- a) 8 bytes
- b) 8 bits
- c) 16 bytes
- d) 124 bytes

[View Answer](#)

Answer: a

The port number is “ephemeral port number”, if the source host is _____

- a) NTP
- b) Echo
- c) Server
- d) Client

[View Answer](#)

Answer: d

“Total length” field in UDP packet header is the length of _____

- a) Only UDP header
- b) Only data
- c) Only checksum
- d) UDP header plus data

[View Answer](#)

Answer: d

Which is the correct expression for the length of UDP datagram?

- a) $\text{UDP length} = \text{IP length} - \text{IP header's length}$
- b) $\text{UDP length} = \text{UDP length} - \text{UDP header's length}$
- c) $\text{UDP length} = \text{IP length} + \text{IP header's length}$
- d) $\text{UDP length} = \text{UDP length} + \text{UDP header's length}$

[View Answer](#)

Answer: a

. The _____ field is used to detect errors over the entire user datagram.

- a) udp header
- b) checksum
- c) source port
- d) destination port

[View Answer](#)

Answer: b